

Triangles — MCQ Worksheet

Mathematics • Chapter 6 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Two figures are congruent if they have:

- (A) Same shape and size
- (B) Same shape, may differ in size
- (C) Same area only
- (D) Same colour

Q2. Two figures are similar if they have:

- (A) Same shape and size
- (B) Same shape, may differ in size
- (C) Same area
- (D) Same perimeter only

Q3. All circles are:

- (A) Congruent
- (B) Similar
- (C) Both
- (D) Neither

Q4. All squares are:

- (A) Congruent
- (B) Similar
- (C) Both
- (D) Neither

Q5. In two similar triangles, corresponding angles are:

- (A) Unequal
- (B) Equal
- (C) Supplementary
- (D) Right angles only

Q6. The Basic Proportionality Theorem (BPT/Thales) states: a line drawn parallel to one side of a triangle divides the other two sides:

- (A) In equal halves
- (B) Proportionally
- (C) At right angle
- (D) Randomly

Q7. The criterion for similarity AAA states that if three angles of one triangle equal three angles of another, the triangles are:

- (A) Congruent
- (B) Similar
- (C) Equilateral
- (D) Right-angled

Q8. The SSS criterion for similarity is when corresponding sides are:

- (A) Equal
- (B) Proportional
- (C) Equal in length only
- (D) Perpendicular

Q9. The SAS criterion for similarity is when one angle is equal and:

- (A) The sides containing that angle are proportional
- (B) All sides are equal
- (C) Hypotenuse equal
- (D) Areas equal

Q10. Pythagoras theorem applies to:

- (A) Equilateral triangles only
- (B) Right-angled triangles
- (C) Isosceles triangles
- (D) Scalene triangles

Q11. In a right triangle with legs 3, 4 the hypotenuse is:

- (A) 5
- (B) 6
- (C) 7
- (D) $\sqrt{7}$

Q12. In a right triangle with legs 5, 12 the hypotenuse is:

- (A) 13
- (B) 17
- (C) 15
- (D) 12.5

Q13. If the sides of a triangle are 6, 8, 10, the triangle is:

- (A) Equilateral
- (B) Right-angled
- (C) Obtuse
- (D) Acute

Q14. Ratio of areas of two similar triangles equals:

- (A) Ratio of corresponding sides
- (B) Square of ratio of corresponding sides

(C) Cube of ratio of sides

(D) No relation

Q15. If ratio of corresponding sides of two similar triangles is 2:3, ratio of areas is:

(A) 2:3

(B) 4:9

(C) 8:27

(D) 6:9

Q16. Two triangles are similar with ratio of areas 16:25. Ratio of corresponding sides is:

(A) 16:25

(B) 4:5

(C) 2:5

(D) 8:25

Q17. In $\triangle ABC$, $DE \parallel BC$; if $AD = 2$, $DB = 3$, $AE = 4$, then $EC =$

(A) 3

(B) 6

(C) 5

(D) 8

Q18. In a triangle ABC , if D and E are midpoints of AB and AC respectively, DE is:

(A) Equal to BC

(B) Half of BC

(C) Equal to AB

(D) Equal to AC

Q19. The triangle formed by midpoints of sides of an equilateral triangle is:

(A) Right-angled

(B) Equilateral

(C) Scalene

(D) Isosceles only

Q20. The triangles in two similar triangles with sides in ratio 1:2 have perimeters in ratio:

(A) 1:2

(B) 1:4

(C) 1:8

(D) 2:1

Q21. In $\triangle ABC \sim \triangle DEF$, $\angle A = 40^\circ$, $\angle B = 60^\circ$. Then $\angle F =$

(A) 40°

(B) 60°

(C) 80°

(D) 120°

Q22. In a right triangle, if a perpendicular is drawn from the right angle to the hypotenuse, the two triangles formed are:

(A) Congruent to original

(B) Similar to original

(C) Equilateral

(D) Right-angled but not similar

Q23. The ratio of areas of two squares whose sides are in ratio 3:5 is:

(A) 3:5

(B) 9:25

(C) 6:10

(D) 27:125

Q24. Two triangles are said to be similar if they satisfy:

(A) AAA only

(B) SSS only

(C) Either AAA, SSS, or SAS criterion

(D) Only equal area

Q25. If the diagonals of a quadrilateral divide each other proportionally, the quadrilateral is a:

(A) Square

(B) Trapezium

(C) Kite

(D) Triangle